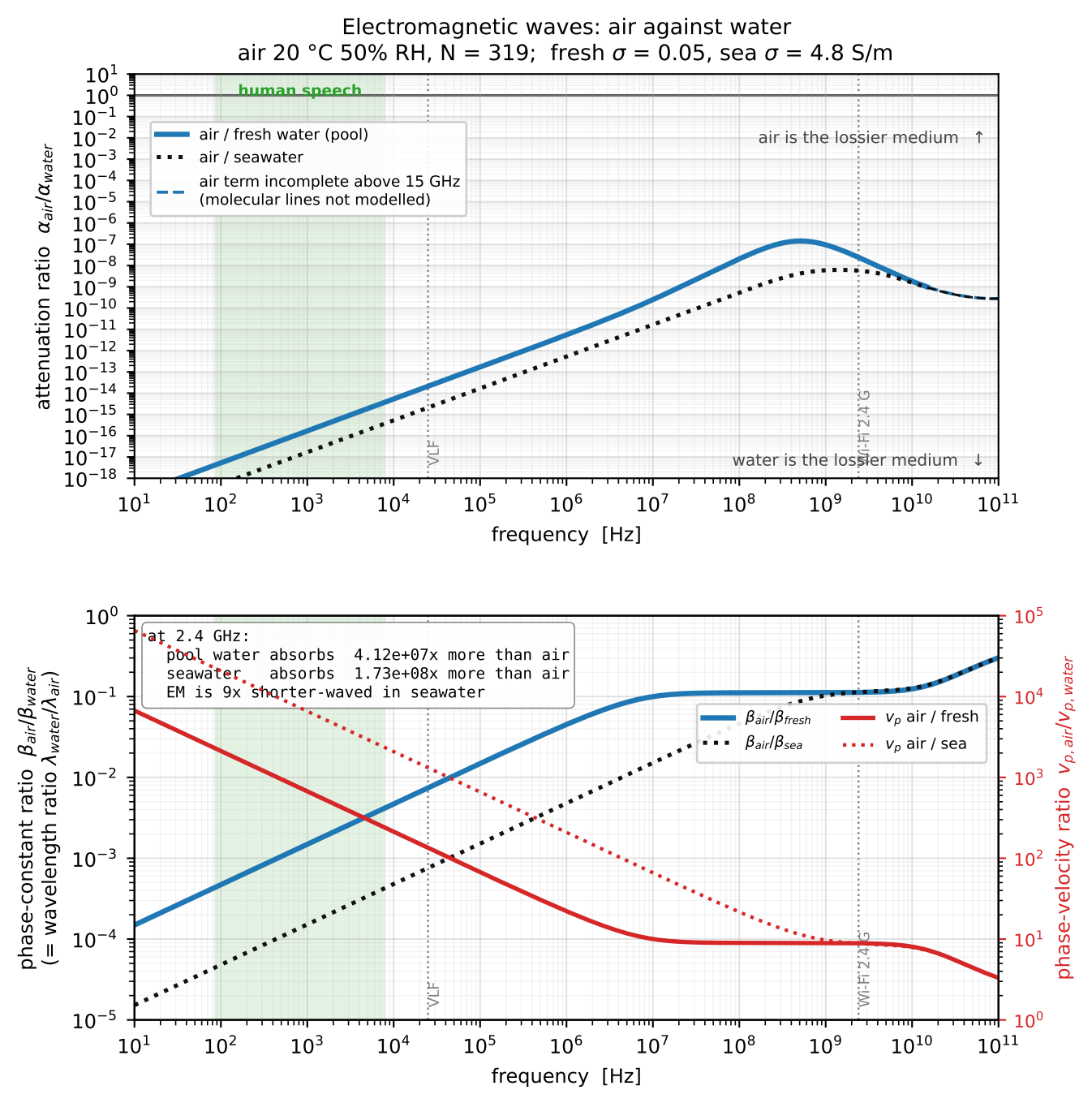

Electromagnetic waves: air against water
air 20 °C 50% RH, N = 319; fresh σ = 0.05, sea σ = 4.8 S/m
human speech
air / fresh water (pool)
air / seawater
air term incomplete above 15 GHz
(molecular lines not modelled)
air is the lossier medium ↑
water is the lossier medium ↓
VLF
Wi-Fi 2.4 G
attenuation ratio $\alpha_{air}/\alpha_{water}$
frequency [Hz]
at 2.4 GHz:
pool water absorbs 4.12e+07x more than air
seawater absorbs 1.73e+08x more than air
EM is 9x shorter-waved in seawater
$\beta_{air}/\beta_{fresh}$
β_{air}/β_{sea}
v_p air / fresh
v_p air / sea
phase-constant ratio $\beta_{air}/\beta_{water}$
(= wavelength ratio $\lambda_{water}/\lambda_{air}$)
phase-velocity ratio $v_{p,air}/v_{p,water}$
frequency [Hz]